

DISTRIBUTION

1. **Simulate 30 rolls with =RANDBETWEEN(1,6). What is the probability of rolling a 3 exactly 5 times? (Hint: Use BINOM.DIST)**

Identify the Binomial Parameters

- Number of trials (n) = 30
- Probability of rolling a 3 on one roll (p) = 1/6
- Number of successes (x) = 5

This follows a Binomial Distribution.

$$P(x = 5) = \binom{30}{5} (1/6)^5 (5/6)^{25} = 0.1705 = 17.05\%$$

Probability of rolling a 3 exactly 5 times in 30 rolls = 0.1705 (approximately 17.05%)

2. **Generate 100 values in Excel using the continuous uniform distribution RAND() and plot a histogram. Describe the shape of the distribution.**

=RAND()

Select the 100 values.

Go to Insert → Statistic Chart → Histogram.

Excel will automatically create a histogram.

The RAND() function generates data from a continuous uniform distribution on the interval [0, 1].

Therefore, the histogram should appear approximately uniform (flat), with bars of similar height across all intervals.

3. **A dataset has a mean of 50 and a standard deviation of 5. What percentage of values lie between 45 and 55 if the data follows a normal distribution?**

Mean (μ) = 50

Standard Deviation (σ) = 5

For $x=45$:

$$z = (45 - 50) / 5 = -1$$

For $x= 55$:

$$Z = (55 - 50) / 5 = 1$$

Since, 45 and 55 are exactly 1 standard deviation below and above the mean:

$$P(45 \leq 55) = 68\%$$

Approximately 68% of the values lie between 45 and 55.

4. **What is the concept of standardization (z-score), and why is it important in data analysis? Explain the formula and how standardization transforms a dataset.**

Standardization is a statistical technique used to transform data so that it has:

- A mean of 0
- A standard deviation of 1

The transformed value is called a z-score.

A z-score indicates how many standard deviations a data point is from the mean.

- $z=0$: the value is exactly at the mean.
- $z=1$: the value is 1 standard deviation above the mean.
- $z=-2$: the value is 2 standard deviations below the mean.

Suppose:

- Mean score = 70
- Standard deviation = 10
- Student's score = 85

Then:

$$z = (85 - 70) / 10 = 1.5$$

This means the student's score is 1.5 standard deviations above the average.

Consider the dataset:

Original Values

40

50

60

Mean = 50, Standard Deviation = 10

Applying the z-score formula:

Original Value	Z-Score
40	-1
50	0
60	1

The transformed dataset becomes:

[-1, 0, 1]

Notice that:

- The mean of the transformed data is 0.
- The standard deviation is 1.
- The relative positions of the data points remain unchanged.
- Standardization converts data into z-scores.
- The formula is:

$$z = (x - \mu) / \sigma$$
- After standardization:
 - Mean = 0
 - Standard deviation = 1
- It helps compare values from different distributions, identify outliers, and prepare data for statistical and machine learning analyses.

5. What is Kurtosis and their type?

Kurtosis is a statistical measure that describes the shape of a distribution, particularly the peakedness and tail heaviness of the data compared to a normal distribution.

It helps us understand:

- How sharply the data is concentrated around the mean.
- How heavy or light the tails of the distribution are.
- The likelihood of extreme values (outliers).

A normal distribution has a kurtosis of 3 (or excess kurtosis = 0).

Types of Kurtosis

1. Mesokurtic
2. Leptokurtic
3. Platykurtic

6. Explain why the uniform distribution is a good model for the outcome of rolling a fair die.

A uniform distribution is a probability distribution in which all outcomes are equally likely to occur.

When a fair six-sided die is rolled, the possible outcomes are:

$\{1,2,3,4,5,6\}$

Since the die is fair, each face has the same chance of appearing:

$$P(1)=P(2)=P(3)=P(4)=P(5)=P(6)=\frac{1}{6}$$

7. Use Excel to compute the probability of getting at least 8 successes in 15 trials with success probability 0.5

Given

- Number of trials (n) = 15
- Probability of success (p) = 0.5
- We want the probability of at least 8 successes:

$$P(X \geq 8)$$

where X follows a binomial distribution.

Excel Formula

Use the cumulative binomial distribution:

$$=1-\text{BINOM.DIST}(7,15,0.5,\text{TRUE})$$

Explanation:

- `BINOM.DIST(7,15,0.5,TRUE)` calculates $P(X \leq 7)$
- Therefore,

$$P(X \geq 8) = 1 - P(X \leq 7)$$

Result

$$P(X \geq 8) \approx 0.5000$$

or

50%

Why is it 50%?

When $n=15$ and $p=0.5$, the binomial distribution is symmetric around its mean:

$$\mu = np = 15 \times 0.5 = 7.5$$

Since 8 is just above the mean and the distribution is perfectly balanced, the probability of getting 8 or more successes is exactly:

0.5

Final Answer

Excel Formula:

$$=1-\text{BINOM.DIST}(7,15,0.5,\text{TRUE})$$

Probability of at least 8 successes:

$$0.5 = 50\%$$

This means there is a 50% chance of obtaining 8 or more successes in 15 trials when the probability of success on each trial is 0.5.

8. How does log transformation help in stabilizing variance and making data more normally distributed?

Log transformation is a data transformation technique in which each value xxx in a dataset is replaced by its logarithm.

A common transformation is:

$$Y = \log(X)$$

It is especially useful when data is right-skewed, contains large outliers, or has non-constant variance.

1. How Log Transformation Stabilizes Variance

In many real-world datasets, the variability increases as the values get larger.

For example:

Income (₹)

20,000

25,000

30,000

2,00,000

5,00,000

The spread among large values is much greater than among small values.
When we apply a logarithm:

Original Value	Log Value (approx.)
20,000	4.30
25,000	4.40
30,000	4.48
2,00,000	5.30
5,00,000	5.70

Here it shows that,

- Large values are compressed more than small values.
- Extreme observations move closer to the rest of the data.
- Variance becomes more uniform across the dataset.

This is called variance stabilization.

.